SIMATS ENGINEERING
ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I ***S. Lakshmi Priya (192524284)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *S. Lakshmi Priya*

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground
Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Answer:

1. Problem Understanding

The knowledge base has one rule (if it rains, the ground becomes wet) and one fact (it is raining). The goal is to prove that the ground is wet. Propositional symbols:

R = It is raining
W = The ground is wet

2. Method Selection

The Resolution Algorithm (proof by refutation) is selected. The goal is negated and added to the knowledge base; if repeated resolution produces the empty clause (□), the negated goal is contradictory, so the original goal is proved. Before resolving, every implication is converted to CNF using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$.

3. Propositional Logic Representation

Statement 1: $R \rightarrow W$
Statement 2: R
Goal: W

4. Conversion to CNF

Clause 1: $R \rightarrow W \equiv \neg R \vee W$
Clause 2: R

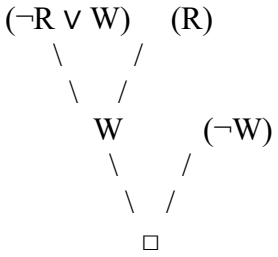
5. Negation of the Goal

$\neg W$ (added as Clause 3)

6. Resolution Steps

Step	Clause	Source
1	¬R ∨ W	Premise (rule)
2	R	Premise (fact)
3	¬W	Negated goal
4	W	Resolve 1 and 2 on R
5	□ (empty clause)	Resolve 3 and 4 on W

Resolution Tree:



7. Interpretation and Final Conclusion

Resolving ¬R ∨ W with R removes R and gives W. Resolving W with the negated goal ¬W gives the empty clause (□), a contradiction. Hence the assumption ¬W is false and the goal W is proved: the ground is wet.

Question 2 – Student Assignment Submission
Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Answer:

1. Problem Understanding

The rule says a student who submits the assignment receives internal marks, and the fact is that Rahul submitted the assignment. The goal is to prove that Rahul receives internal marks. Propositional symbols:

S = Rahul submitted the assignment
M = Rahul receives internal marks

2. Method Selection

The Resolution Algorithm (proof by refutation) is selected. The goal is negated and added to the knowledge base; if repeated resolution produces the empty clause (□), the negated goal is contradictory, so the original goal is proved. Before resolving, every implication is converted to CNF using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$.

3. Propositional Logic Representation

Statement 1: $S \rightarrow M$
Statement 2: S
Goal: M

4. Conversion to CNF

Clause 1: $S \rightarrow M \equiv \neg S \vee M$
Clause 2: S

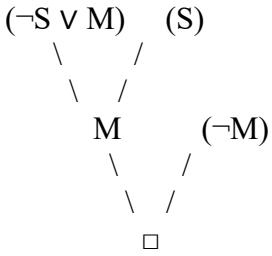
5. Negation of the Goal

$\neg M$ (added as Clause 3)

6. Resolution Steps

Step	Clause	Source
1	$\neg S \vee M$	Premise (rule)
2	S	Premise (fact)
3	$\neg M$	Negated goal
4	M	Resolve 1 and 2 on S
5	□ (empty clause)	Resolve 3 and 4 on M

Resolution Tree:



7. Interpretation and Final Conclusion

The empty clause (□) is derived, so the set {$\neg S \vee M$, S, $\neg M$} is unsatisfiable. Therefore the goal is proved: Rahul receives internal marks.

Question 3 – Library Membership

Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Answer:

1. Problem Understanding

The rule says a library member can borrow books, and the fact is that Priya is a library member. The goal is to prove that Priya can borrow books. Propositional symbols:

L = Priya is a library member
B = Priya can borrow books

2. Method Selection

The Resolution Algorithm (proof by refutation) is selected. The goal is negated and added to the knowledge base; if repeated resolution produces the empty clause (□), the negated goal is contradictory, so the original goal is proved. Before resolving, every implication is converted to CNF using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$.

3. Propositional Logic Representation

Statement 1: $L \rightarrow B$
Statement 2: L
Goal: B

4. Conversion to CNF

Clause 1: $L \rightarrow B \equiv \neg L \vee B$
Clause 2: L

5. Negation of the Goal

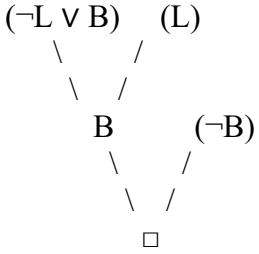
$\neg B$ (added as Clause 3)

6. Resolution Steps

Step	Clause	Source
1	$\neg L \vee B$	Premise (rule)
2	L	Premise (fact)
3	$\neg B$	Negated goal
4	B	Resolve 1 and 2 on L
5	□ (empty clause)	Resolve 3 and 4 on B

Other pairs: Clauses 1 and 3 give $\neg L$, which resolved with Clause 2 (L) gives □ again. No other new clauses can be generated, so the process stops once □ is derived.

Resolution Tree:



7. Interpretation and Final Conclusion

Resolution is applied until no new clauses can be generated, and it ends with the empty clause (□). The negated goal contradicts the knowledge base, so the goal is proved: Priya can borrow books.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Answer:

1. Problem Understanding

The rule says a student who clears the aptitude test is eligible for placement, and the fact is that Arun cleared the aptitude test. The goal is to prove that Arun is eligible for placement. Propositional symbols:

A = Arun cleared the aptitude test
E = Arun is eligible for placement

2. Method Selection

The Resolution Algorithm (proof by refutation) is selected. The goal is negated and added to the knowledge base; if repeated resolution produces the empty clause (□), the negated goal is contradictory, so the original goal is proved. Before resolving, every implication is converted to CNF using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$.

3. Propositional Logic Representation

Statement 1: $A \rightarrow E$
Statement 2: A
Goal: E

4. Conversion to CNF

Clause 1: $A \rightarrow E \equiv \neg A \vee E$
Clause 2: A

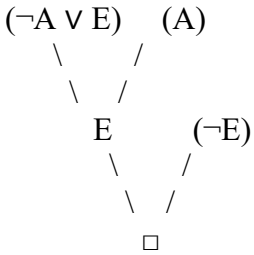
5. Negation of the Goal

$\neg E$ (added as Clause 3)

6. Resolution Steps

Step	Clause	Source
1	$\neg A \vee E$	Premise (rule)
2	A	Premise (fact)
3	$\neg E$	Negated goal
4	E	Resolve 1 and 2 on A
5	□ (empty clause)	Resolve 3 and 4 on E

Resolution Tree:



7. Interpretation and Final Conclusion

The resolution tree ends in the empty clause (□), which shows the negated goal is inconsistent with the knowledge base. Hence the goal is proved: Arun is eligible for placement.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

Answer:

1. Problem Understanding

The knowledge base has two rules (correct password implies authenticated, and authenticated implies access granted) and one fact (the correct password was entered). The goal is to prove that the user is granted access. Propositional symbols:

P = The user enters the correct password
A = The user is authenticated
G = The user is granted access

2. Method Selection

The Resolution Algorithm (proof by refutation) is selected. The goal is negated and added to the knowledge base; if repeated resolution produces the empty clause (□), the negated goal is contradictory, so the original goal is proved. Before resolving, every implication is converted to CNF using the equivalence $P \rightarrow Q \equiv \neg P \vee Q$.

3. Propositional Logic Representation

Statement 1: $P \rightarrow A$
Statement 2: $A \rightarrow G$
Statement 3: P
Goal: G

4. Conversion to CNF

Clause 1: $P \rightarrow A \equiv \neg P \vee A$

Clause 2: $A \rightarrow G \equiv \neg A \vee G$

Clause 3: P

5. Negation of the Goal

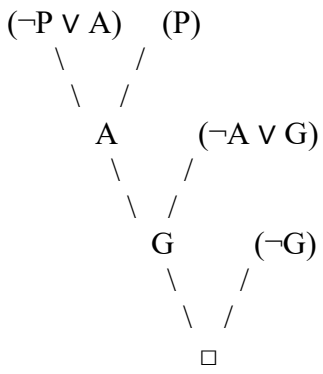
$\neg G$ (added as Clause 4)

6. Resolution Steps

Step	Clause	Source
1	$\neg P \vee A$	Premise (rule 1)
2	$\neg A \vee G$	Premise (rule 2)
3	P	Premise (fact)
4	$\neg G$	Negated goal
5	A	Resolve 1 and 3 on P
6	G	Resolve 2 and 5 on A
7	□ (empty clause)	Resolve 4 and 6 on G

Step 5: Clause 1 contains $\neg P$ and Clause 3 is P, so P is cancelled and A remains (the user is authenticated). Step 6: Clause 2 contains $\neg A$ and Step 5 gives A, so A is cancelled and G remains (access is granted). Step 7: G contradicts the negated goal $\neg G$, giving the empty clause.

Resolution Tree:



7. Interpretation and Final Conclusion

The empty clause (□) is derived, so the knowledge base together with $\neg G$ is unsatisfiable. Therefore the goal is proved: the user is granted access. The two rules chain together, so a correct password leads to authentication and then to access.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation